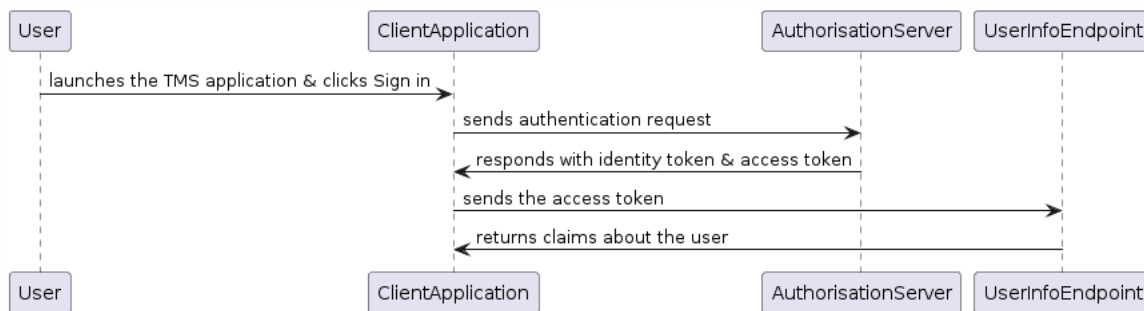


Access Control and Security

When selecting the best authentication solution for the Task Management System (TMS), the project group considered the ones that can provide an identity layer and enable secure authentication. OpenID Connect is the one that satisfies this requirement since it is an authentication protocol based on the OAuth 2.0 authorisation framework (OpenID, n.d.). It verifies the identity of users during the authentication process performed by `:AuthorisationServer`. Once the user has launched the task management system, they type their username and password and then click on the Sign in button. The client application sends a request to `:AuthorisationServer`, and it responds with an identity token and access token. The application uses the access token at the `:AuthorisationServer`'s user info endpoint retrieving claims about the user.

Figure 1

The UML Sequence Diagram of the Interactions between the User and AuthorisationServer



Source: Own results.

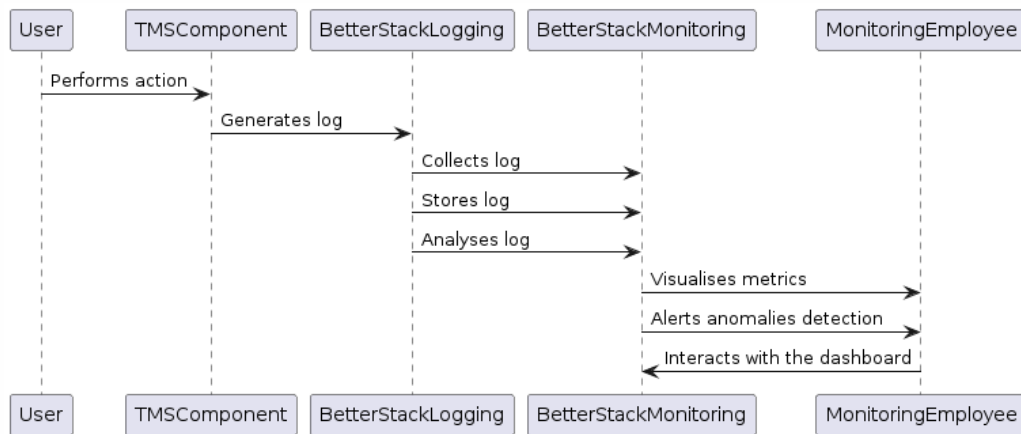
Global Software Control

Global software control for a task management system requires designing a control system that would monitor the entire system's operations, ensuring consistency and reliability. This control system should include logging and monitoring services across different systems' components.

When selecting a log monitoring service, the project group considered that this service should implement log collection, storing, analysis, and alerting and provide a dashboard. Better Stack meets these requirements since it can collect logs from all servers and databases and parse them into structured JSON format that can be queried. As this service focuses on system reliability and performance, it analyses logs for errors and unusual patterns and identifies trends by visualising metrics. It also provides a dashboard where employees are able to collaborate and resolve problems together. Additionally, the service can alert the TMS monitoring employees about anomaly detection in logs.

Figure 2

The UML Sequence Diagram of the Interactions between the User and the Log Monitoring Service Better Stack



Source: Own results.